

Время на чтение: интенсивные 7:18

1.

"Good afternoon. My name is Ilya Bersenev, and today I would like to present my bachelor's thesis on fast AV1 video decoding. This work was carried out in collaboration with LLC COMEXP, whose product TAPE enables video-by-video search. The core problem we address is that decoding currently dominates the entire pipeline."

2.

This study addresses a pipeline bottleneck observed in the current system. Komexp employs a patented video indexer—functionally analogous to a unidirectional encoder—that traverses decoded frames to generate a compact index file. However, the indexing throughput exceeds decoding throughput by more than 20, creating a performance imbalance.

The project aims to create a system that achieves the highest possible decoding speed for AV1 video. Experiments with the TAPE framework indicate negligible performance variation between standard-resolution and ultra-low-resolution streams, justifying the selection of a 64-pixel resolution output format for benchmarking.

3.

The object of study is the reference AV1 decoder Libaom; the subject is its single-threaded performance at a 64-pixel output resolution. The research aim is to evaluate and implement algorithmic optimisations without parallelism or GPU offloading. We hypothesise that targeted changes to the most expensive decoder stages can cut total processing time by 30 to 50 percent while keeping video quality above a usable threshold

4.

To date, we have defined functional and non-functional requirements; analyzed the decoder specification; pinpointed performance bottlenecks in the av1 standard; identified acceleration techniques; selected an appropriate technology stack; and drafted both naïve and optimized architectures. The forthcoming tasks are implementation development and comprehensive testing.

5.

The literature review allowed us to pinpoint three major cost centres in AV1 — deblocking, CDEF filtering and sub-pixel interpolation. Let's take a look at them

6.

We begin with interpolation. The article by Chen et al explains that interpolation in AV1 is one of the most difficult calculations which employs an 8-tap filter with eight coefficient values, in contrast to H.264's simpler scheme. Even when data-fetch latency is ignored, a single-pixel interpolation requires a minimum of 48 CPU cycles, which is computationally expensive.

7.

The same paper also distinguishes the deblocking filter, which mitigates block artifacts by processing the edges of each transform block. Although essential for visual quality at higher resolutions, its computational cost is disproportionate at the target 64-pixel output, where the visual benefit is negligible.

8.

The Constrained Directional Enhancement Filter (CDEF) is another post-processing stage that operates on block boundaries, applying a set of eight directional filters to suppress ringing and edge artifacts. Similar to deblocking, its impact diminishes at very low resolutions.

9.

Lookup tables store pre-computed approximations (e.g., cosine values) that are accessed via an array index serving as the function argument and may help in optimizing. The slide illustrates replacing on-the-fly cosine calculations with a table of rounded values, thereby reducing arithmetic overhead. A paper by Yi and Song, however, states that such operations may be harmful for the performance due to the cache misses.

10.

We now turn to the methodological base of the study. During the literature review we distinguished four pillars.

First one is to replace lookup tables with algebraic expressions. According to the Yi and Song study, that may increase performance of certain operations.

Next one is to use magic constant division, as stated in the article by of the IEEE (I triple e) to speed up interpolation.

Third is to skip costly post-processing steps when target output is less than or equal to 64px

11.

The project will be evaluated using these metrics:

end-to-end latency from video upload to completion of decoding;

pure decoder execution time;

PSNR;

per-frame luma-difference quality metric (original vs. decoded);

stability and reproducibility of the applied optimizations;

and response time under load-testing conditions.

12.

plan of the practice part is implementing the naïve and optimized realizations and evaluating them,

13.

Expected results: at least 20 % reduction of total time from video upload to decoding; at least 30 % acceleration of video decoding time; acceptable video degradation is characterized by a PSNR metric above 15 decibels and the absence of artifacts affecting TApe operation. These metrics will be identified in comparison with the naive implementation, which will be described later.

14.

Thank you for your attention — I am happy to take your questions.